Jasper Buntinx

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EDUCATION

Texas A&M University | *Bachelor of Science, Mechanical Engineering* (GPA: 3.31 / 4.0) **College Station, TX** **Aug 2023 - 2027**

WORK EXPERIENCE

Rochester Sensors (Amphenol Subsidiary) | *Mechanical Engineering Intern* **May 2026 - Aug 2026**

- Programmed CNC machining workflows in CamWorks for 15+ production parts and dedicated multi-part fixtures, replacing inherited pattern-projected toolpaths to cut cycle times 70–80% (182 → 34 min on highest-volume part) while holding all GD&T requirements.
- Resolved tolerance stack-up and assembly fit defects on in-development products in SolidWorks, validating designs by machining prototypes to released dimensions and correcting a source-design misalignment before mass production.
- Executed environmental qualification testing - accelerated thermal cycling (−40 to +125 °C), salt ingress, and vibration - on sensor assemblies for pre-launch client products, demonstrating reliability beyond 50-year equivalent field life against a 30-year design requirement.
- Programmed Universal Robots arm for vision-guided bin picking, using Keyence 3D scanning to grip randomly oriented parts from bulk bins and sort them at 15–30 parts/min, automating the first station of a phased plan to redeploy operator hours.

Waterloo Greenway Conservancy (Austin Creek Show) | *Mechanical Engineering Intern* **May 2022 - Jan 2023**

- Derived four unique SolidWorks assemblies from Odonata LLC artist samples, enabling production of large metal dragonfly structures each with 14-ft+ wingspans.
- Led fabrication of prototyped and full-scale dragonfly structures by laser-cutting eight unique wing designs via Fusion 360 CAM, metal frame fabrication, welding 200+ joints, soldering wiring for four LED displays, and producing water-resistant electronics packaging.
- Reduced effective resistant area and cut peak wind/flood loads by 30% through swapping hard plastic shells for a microporous ePTFE membrane.
- Reinforced against peak bending-moment forces via wing reorientation, steel base widening, and implementing 300lb ballast per structure while verifying/modeling in SolidWorks Simulation.

Petrobras | *Mechanical Engineering Intern* **Jun 2025 - Aug 2025**

Optimized CUDA-based Lattice Boltzmann (LBM) CFD solvers using AI-assisted kernel profiling in NVIDIA Nsight Compute, resolving memory bottlenecks to push throughput past published open-source LBM benchmarks:

- Achieved 400%+ throughput gain (MLUPS) via FP64→FP32 migration and memory-access restructuring, validated against solver tolerances.
- Authored a Python mesh-generation script for fine-to-coarse grid interfaces, raising the stable peak Reynolds number 33% with no throughput loss.
- Diagnosed uncoalesced global memory access via Nsight Compute workload reports, remapping thread-to-data access for 60% better warp efficiency.
- Cut peak shared memory 35% (56KB→36KB) after identifying redundant data staging and enabling aggressive reuse across thread blocks.

PROJECTS

Autonomous Laundry-Picking Cable Claw (Prototype) **May 2026 - Present**

- Prototyped a room-scale, 4-winch cable-driven robot that detects laundry with an overhead camera (YOLO, OpenCV, Python) and delivers 0-1.5 kg payloads to hamper.
- Engineered the full electrical system: fused 12 V power with an emergency kill switch feeding Arduino/CNC-shield stepper control (A4988) with fail-safe NC limit-switch homing, plus a separate LiPo-powered ESP32 Wi-Fi end-effector driving the tendon gripper through a buck-regulated servo.
- Modeled 10+ parts in SolidWorks for FDM printing, including a five-finger tendon-driven TPU gripper engineered to grasp wide array of fabrics/textures.
- Derived inverse kinematics for a four-cable ceiling robot, sizing NEMA 17 steppers with 2.6 FOS for worst-case tension by validating 4kg weight.

Automatic Retractable RFID Bike Lock (Prototype) **Dec 2025 - Present**

- Designed a frame-mounted, NFC-unlocked bike lock with a retractable steel cable: 15-part 3D-printed clamshell assembly integrating a solenoid latch, PN532 RFID reader, Arduino Nano, Li-ion power system, and a spring-loaded cable spool.
- Manufactured 3D-printed PETG prototype housing with TPU compliant liner fitting Ø27–46 mm frames; now iterating toward a CNC-milled stainless version with DFM constraints designed in from the start.

PLA Filament Recycler | *Aggies Create (Ecofil Team)* **Aug 2025 - May 2026**

- Designed three-stage filament recycler in SolidWorks, applying GD&T to control mating fits and DFM to constrain every feature for feasible fabrication in-house.
- Held extruded filament diameter to ±0.10 mm using closed-loop control, a linear Hall-effect sensor driving a stepper puller that corrects draw rate mid-run.
- Fabricated functional prototypes of the shredder, dehydrator, and extruder through 3D printing (OrcaSlicer), CNC milling (SolidWorks CAM), laser cutting (Lightburn), soldering, and MIG welding, providing working models for performance testing and design iteration.
- Utilize Arduino controllers, high-voltage AC motors, linear Hall-effect sensors, MOSFETs, PTC heating elements, thermistors, stepper motors, fans, and hygrometers to enable real-time monitoring and control of the recycling process.

VEX U Competition Robots | *Aggie Robotics (Team WHOOP5)* **Aug 2023 - May 2024**

- Secured 1st place at 2024 VEX U OVER UNDER State Championship and placement at the World Championship, contributing through AutoCAD schematic redesign and iterative development of an extendable friction launcher to retain, launch, or push Tri-balls as needed.

SKILLS

Software: SolidWorks, CSWA Certified, AutoCAD, OnShape, SolidWorks Simulation, NX, ANSYS, MATLAB, ParaView, Nvidia Nsight Compute, MS Office Suite, SolidWorks CAM, OrcaSlicer, Fusion 360 CAM, Lightburn, Mastercam
Programming: CUDA, C++, Python, Java, HTML, Arduino/Embedded C
AI Tooling: Claude Code, Cursor, API Integration, Multi-Agent Orchestration, Token Optimization
Fabrication: GD&T, FEA, 3D printing, Woodworking, Metalworking, Soldering, MIG Welding, Injection Molding, CNC, Hand tools

EXTRACURRICULAR

One Army Men's Organization | *Special Events Logistics Chair* **Aug 2025 - Present**

- Led large event construction on $30K+ annual budget, taking builds from design/mockup through BOM and fabrication w/ help from 80+ members for final assembly.
- Work year-round on sponsorship acquisition, obstacle construction, and promotion for Gladiator Dash, a 501(c)(3) 5K mud run raising $180K+ annually.

Freshmen Leaders in Progress (FLIP) | *Service Chair* **Aug 2023 - May 2025**

- Responsible for the recruitment of 56 freshmen, integration of new members into college life, and coordination of volunteer events with entities including local food pantries, schools, and Habitat for Humanity.